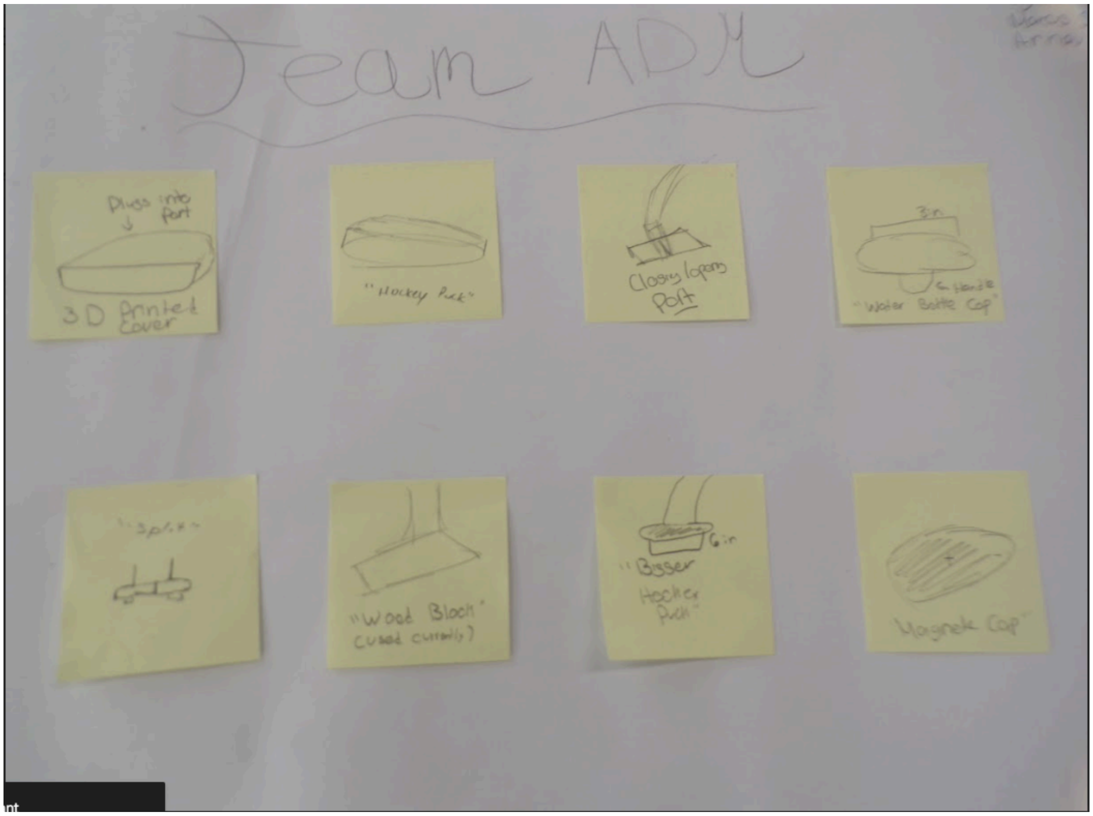
Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

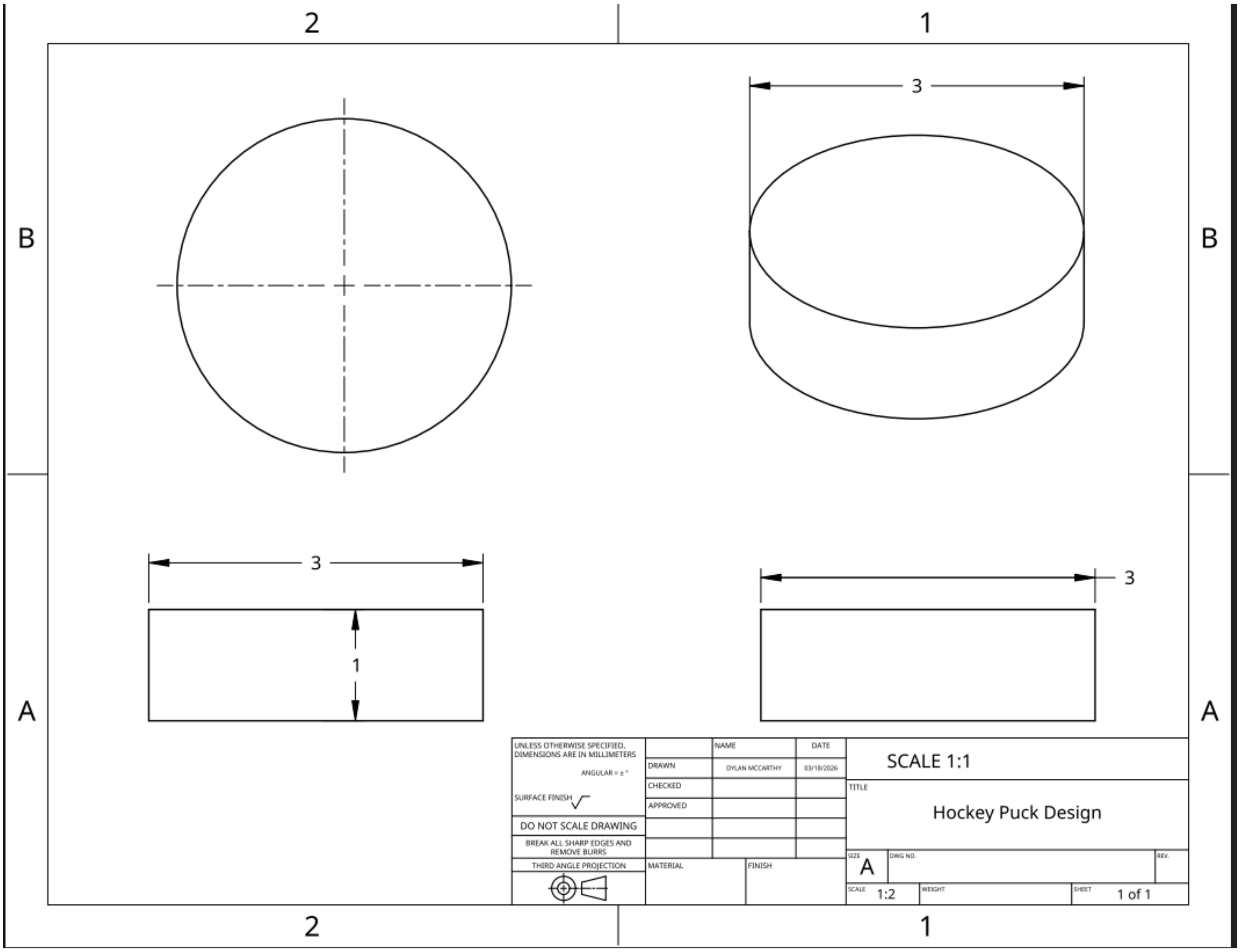


Design Idea #1 (Hockey Puck Idea)

Date: 3/18/26
Description: A 3 inch circular diameter hockey puck shape that will be approximately 1 inch

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

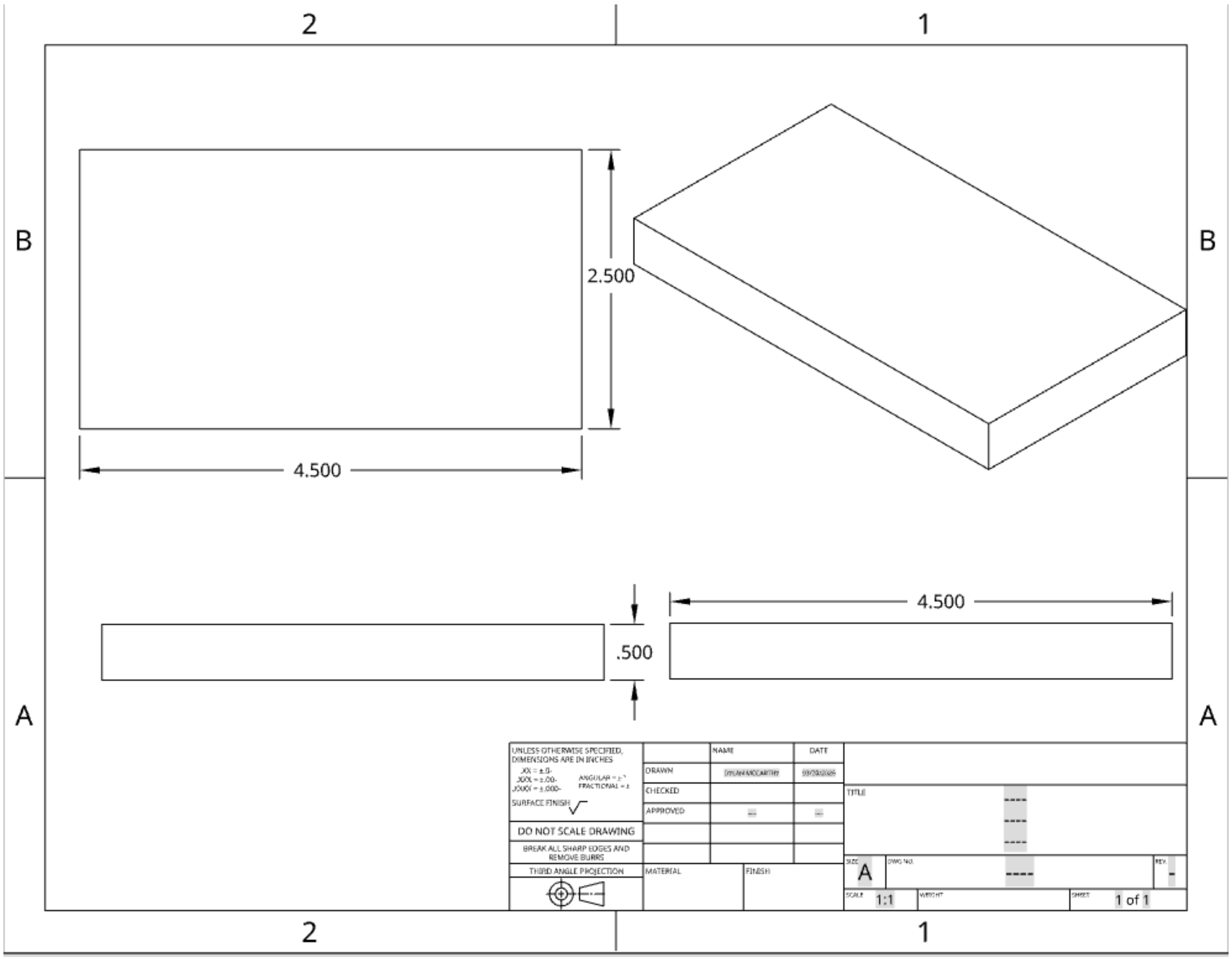
thick and block airflow in an open port



Design Idea #2 (Wooden Block)

Date: 3/20/26

Description: A wooden block cut to 4.5 inches by 2.5 inches and .5 thickness used to block airflow. This design has been in use as of current by Mr. D and it seems to work but not an ideal solution.

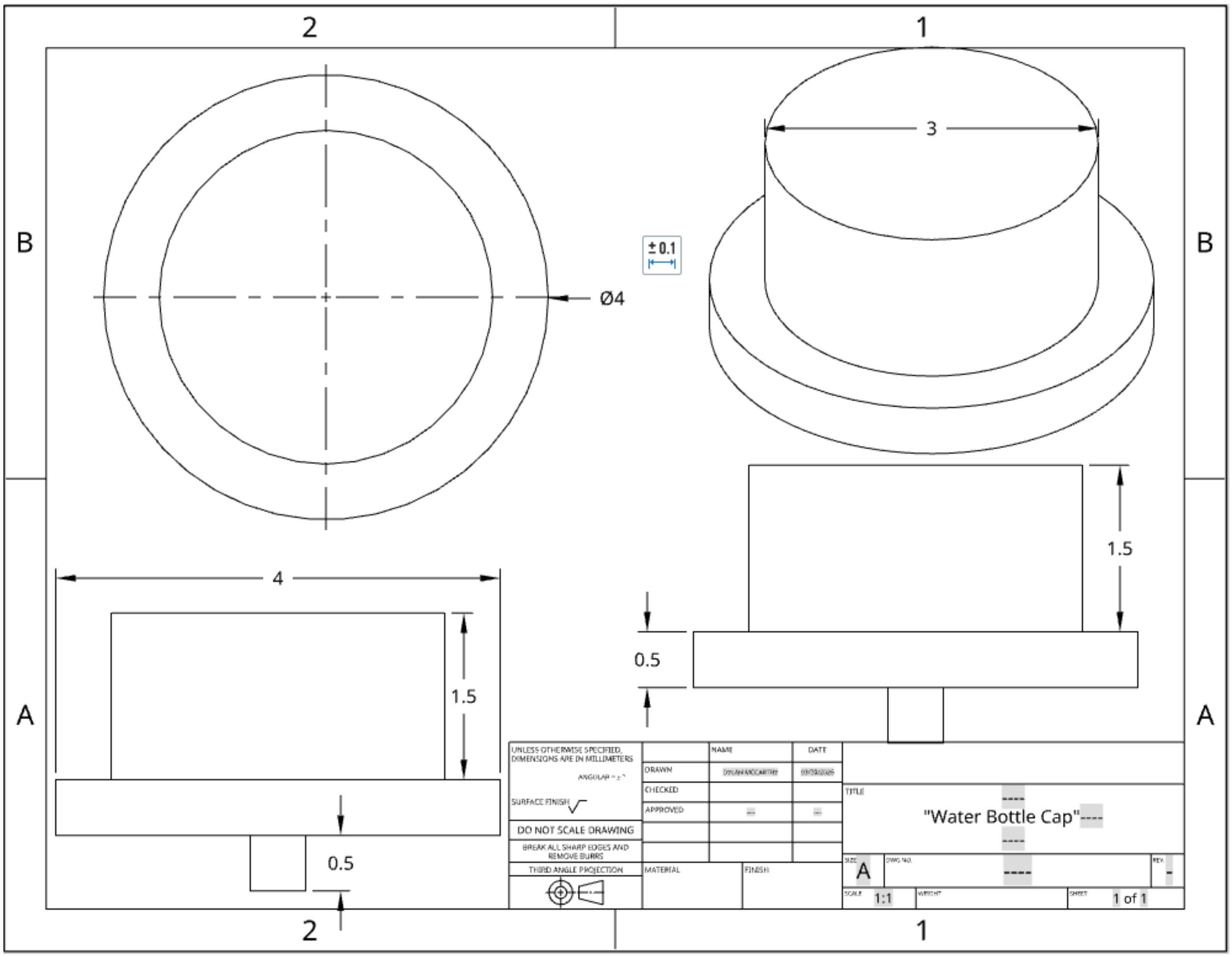


Design Idea #3 (Fitted Cap)

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

Date:3/20/26

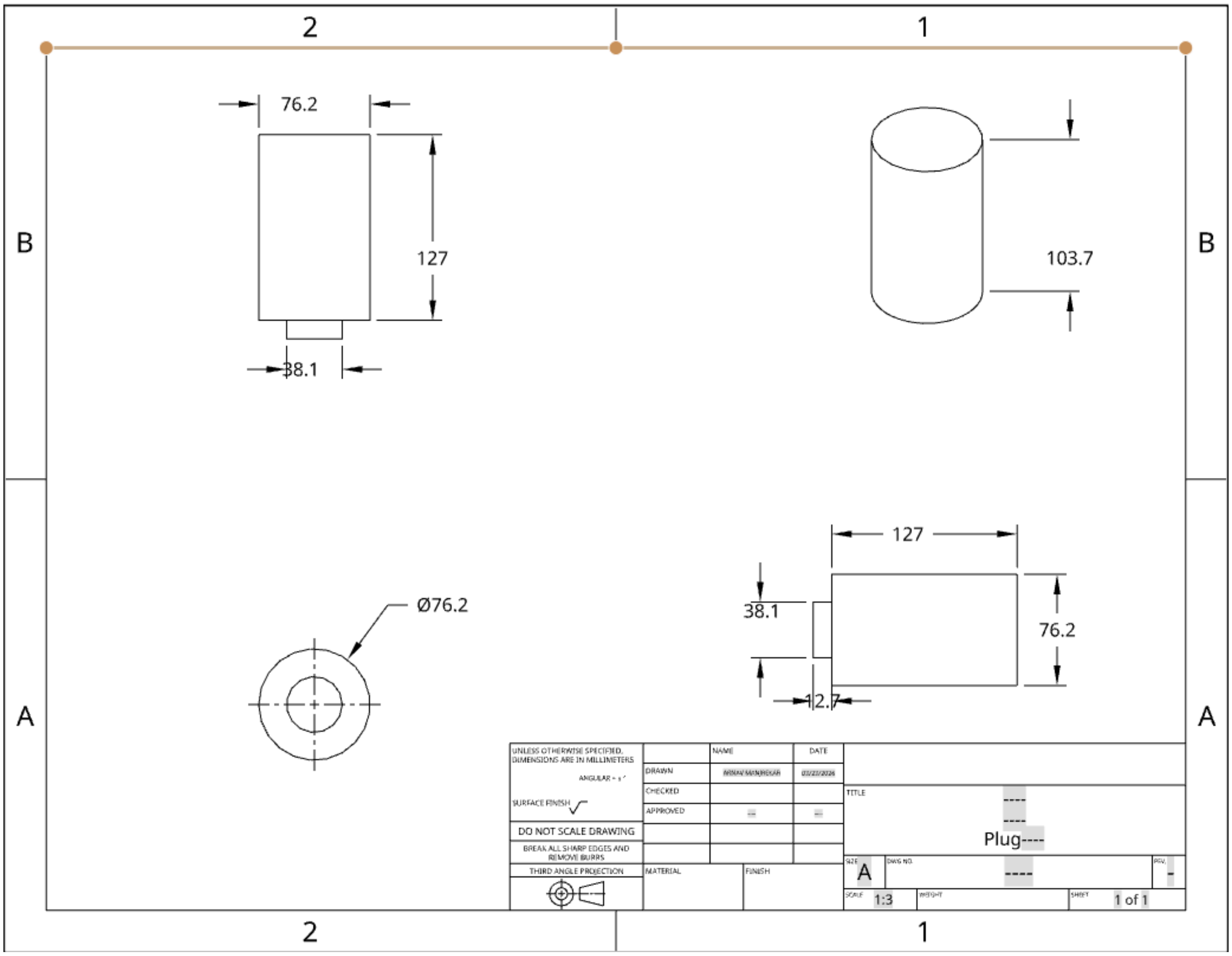
Description: A cap that plugs into the 3 in port, has a handle to insert and remove.



Design Idea #4 (Plug)

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

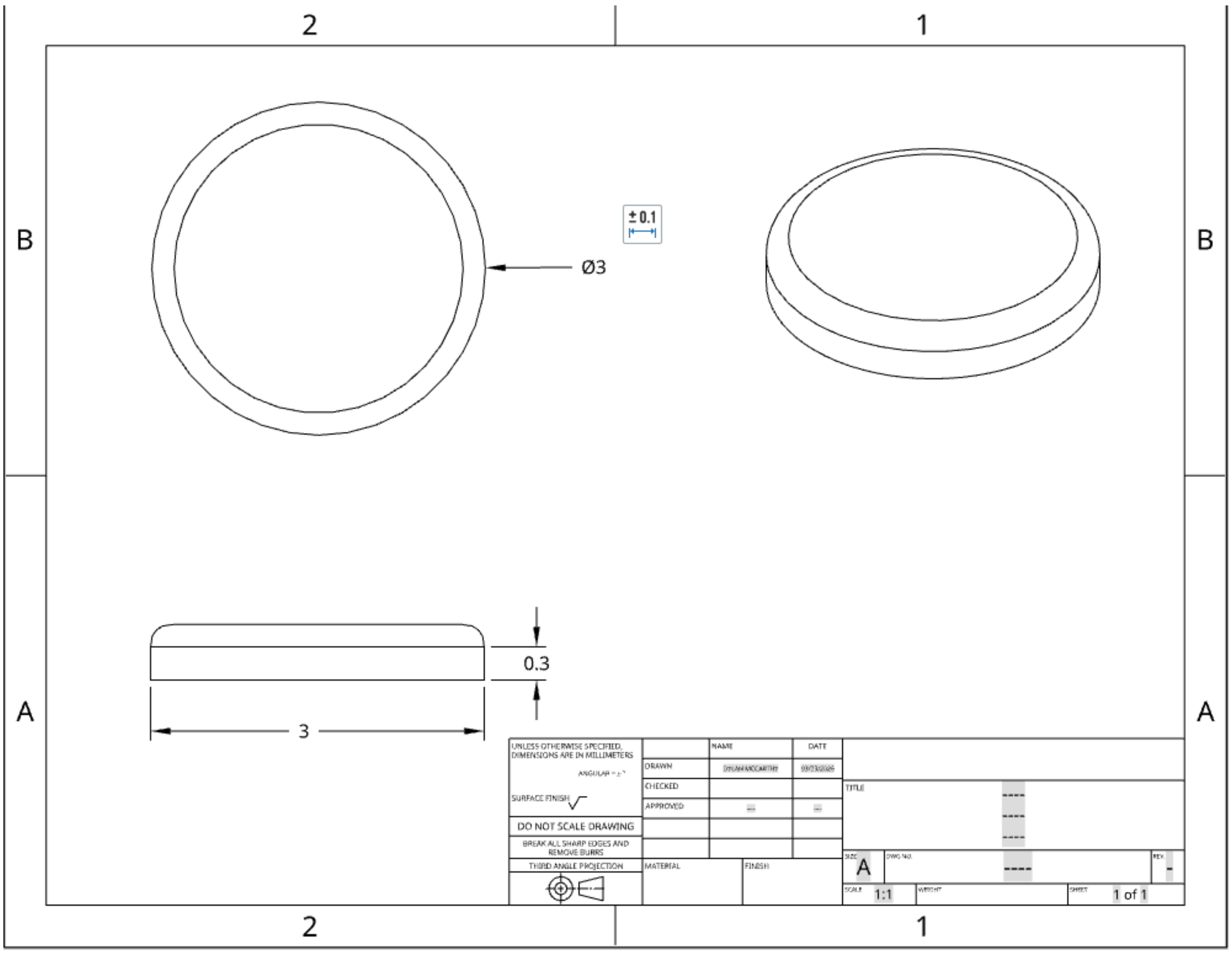
Date: 3/23/26
Description: A longer block with a plug at the bottom that fits into the pipe and blocks air



Design Idea #5 (Simple Cover)

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

Date: 3/23/26
Description: A simple cover with a fillet and shell to cover the the 3 inch diameter that is on the port



Design Idea Comparison and Selection *(D1, D2)*

engineering 4 us all

Student Handout

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

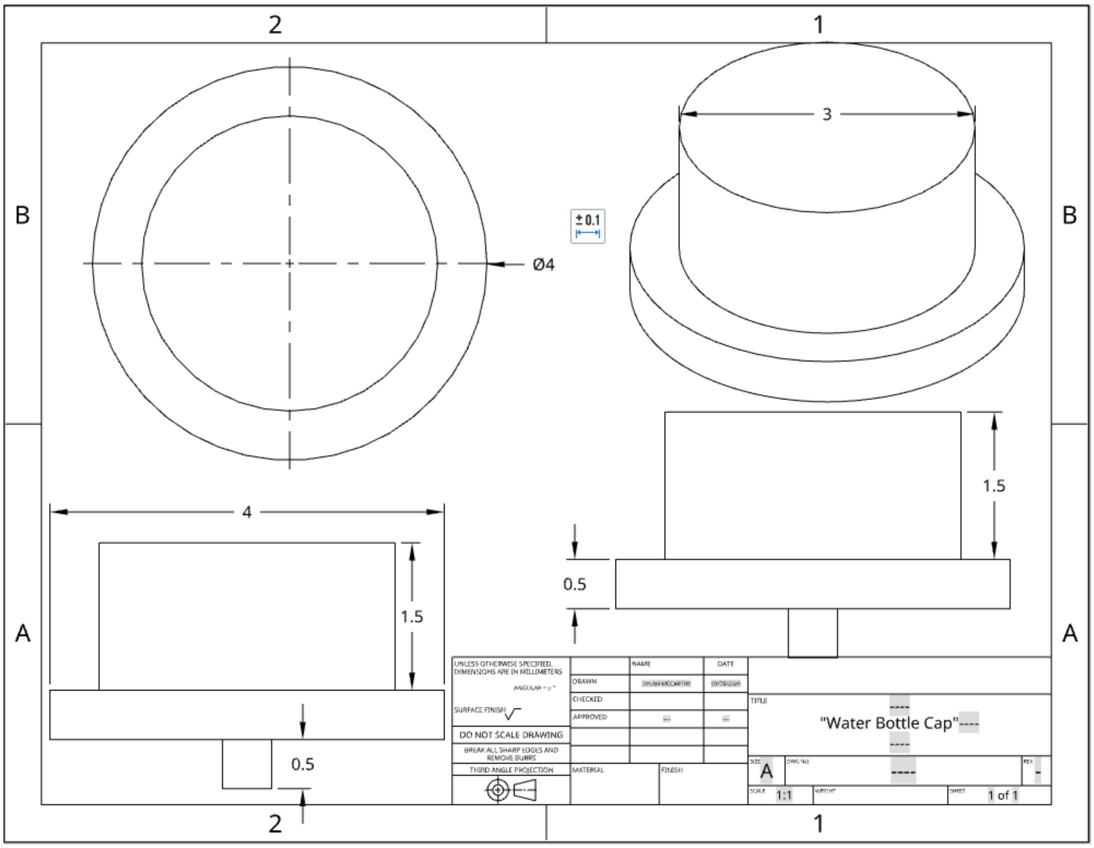
Function	*Weight*	*Design Idea #1 Hockey Puck Idea*	*Design Idea #2 Wooden Block*	*Design Idea #3 Fitted Bottle Cap Idea*	*Design Idea #4 Plug*	*Design idea #5 Simple Cap*
Strength	30%	5	4	2	2	2
Visual Appearance	15%	1	1	3	2	4
Ease of Manufacturing	20%	2	5	4	5	5
Fit/Functionality	25%	4	3	5	3	3
Costs	10%	3	5	4	4	4
Score (1 worst, 5 Best)		3.35	3.23	3.5	3.05	3.35

Design Blueprint *(D4)*

Design Blueprint #3 Fitted Bottle Cap

Date:3/24/26

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design



Manufacturing Methods: 3D Print

Step 1: Design on Onshape

Step 2: 3D PrintSupply List and Costs

Item	Cost each	# needed	Total
Computer	$0	1	$0
PLA	$20	⅛ roll	$0.25
Total Cost $0.25 / cap			

Name: Team ADM **Date:** 3/17/2026 **Class:** Engineer Design

Plan of Action *(D3)*